

# Michael O'Loughlin

## WA Bee Atlas Collection and Identification Report - 2024

### 1 Your 2024 Collections

Michael O'Loughlin caught 109 bees across 1 county from April 11, 2024 to October 10, 2024 representing 37 unique taxa, including 34 unique species.

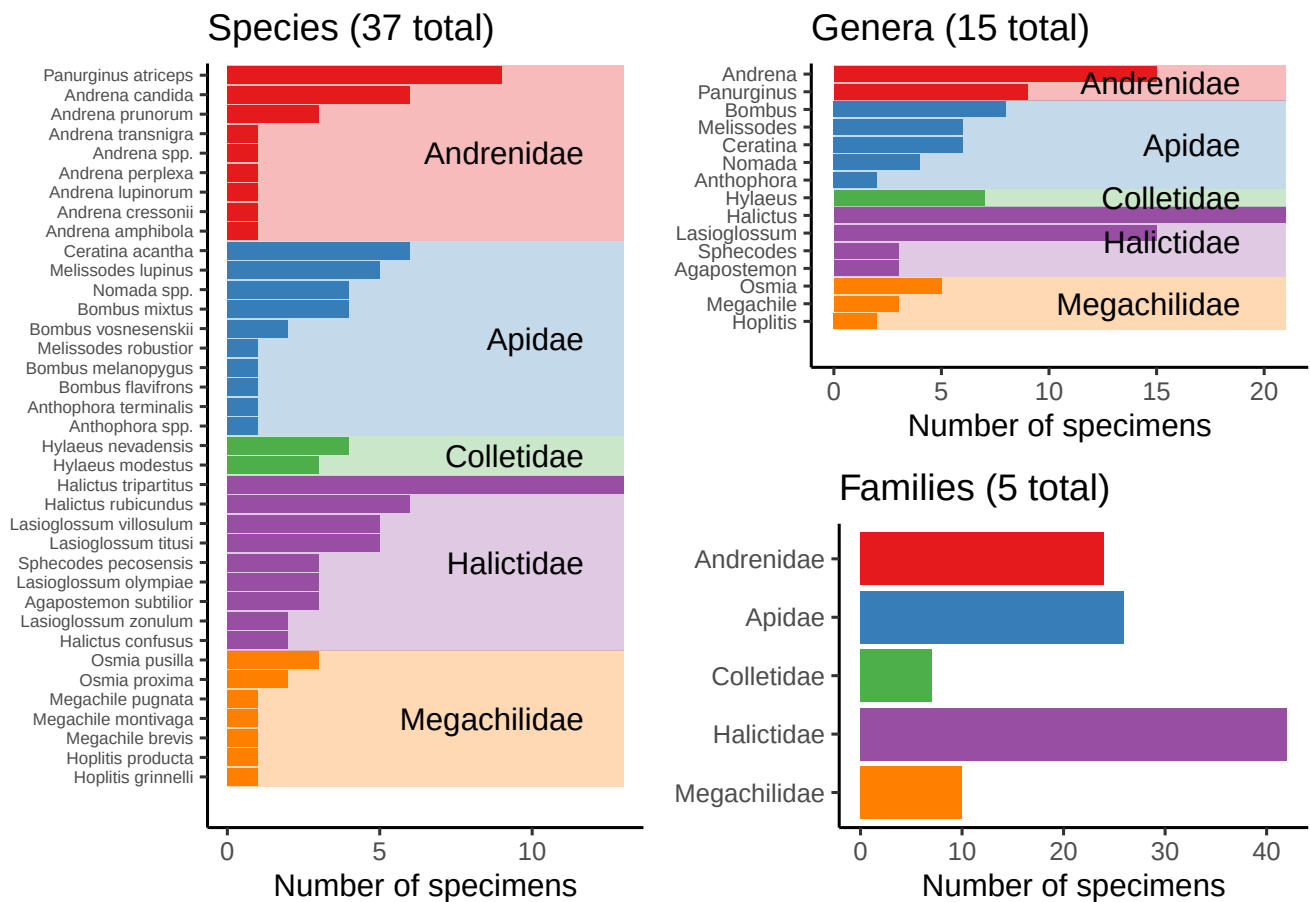


Figure 1: Bees caught by Michael O'Loughlin, broken down by species, genus, and family.

# 2 All Your Collections

Michael O'Loughlin caught 259 bees across 1 county from July 21, 2023 to August 13, 2025, representing 60 unique taxa, including 52 unique species. Michael O'Loughlin also caught the only *Sphecodes arvensiformis* in the collection!

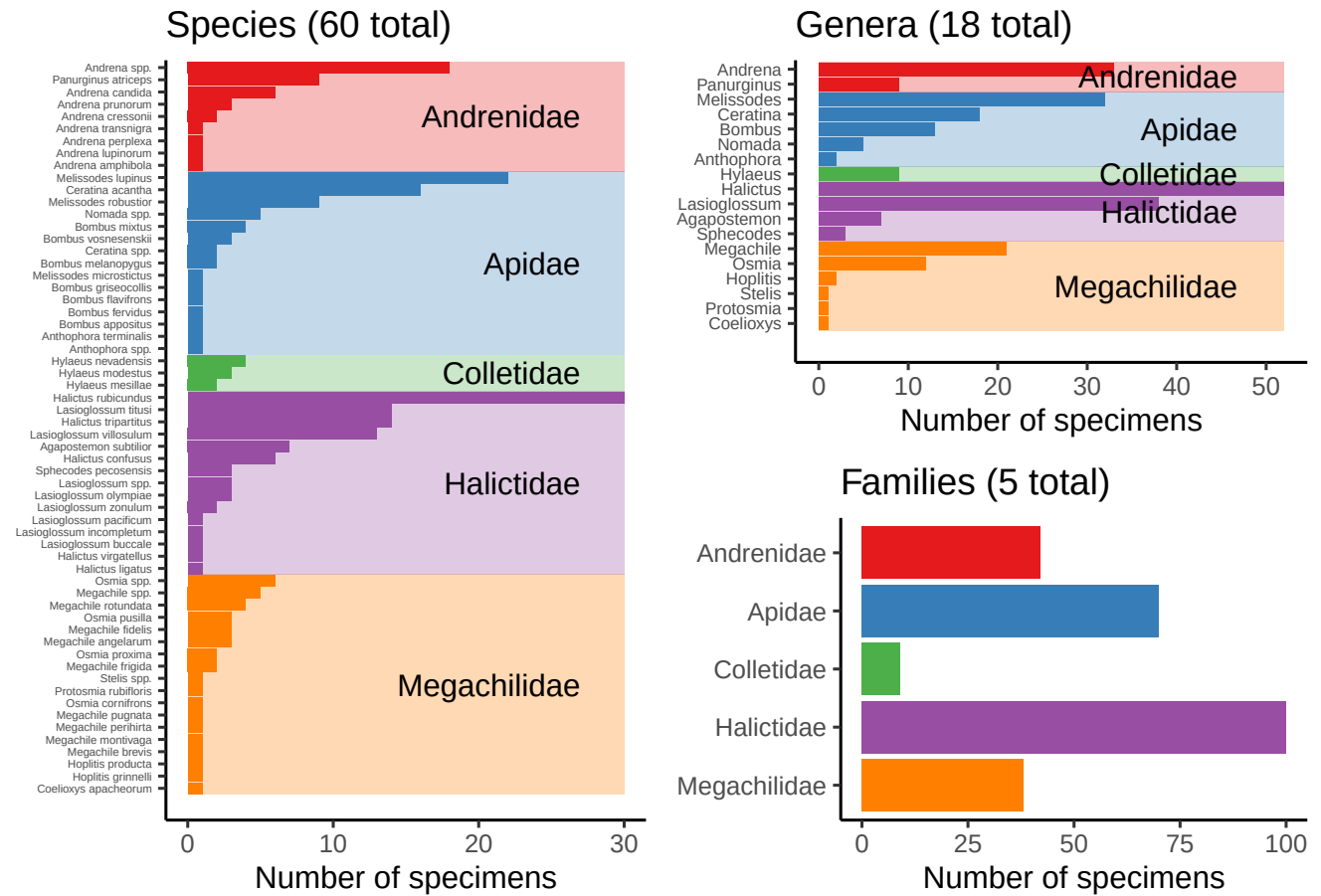


Figure 2: Bees caught by Michael O'Loughlin, broken down by species, genus, and family.

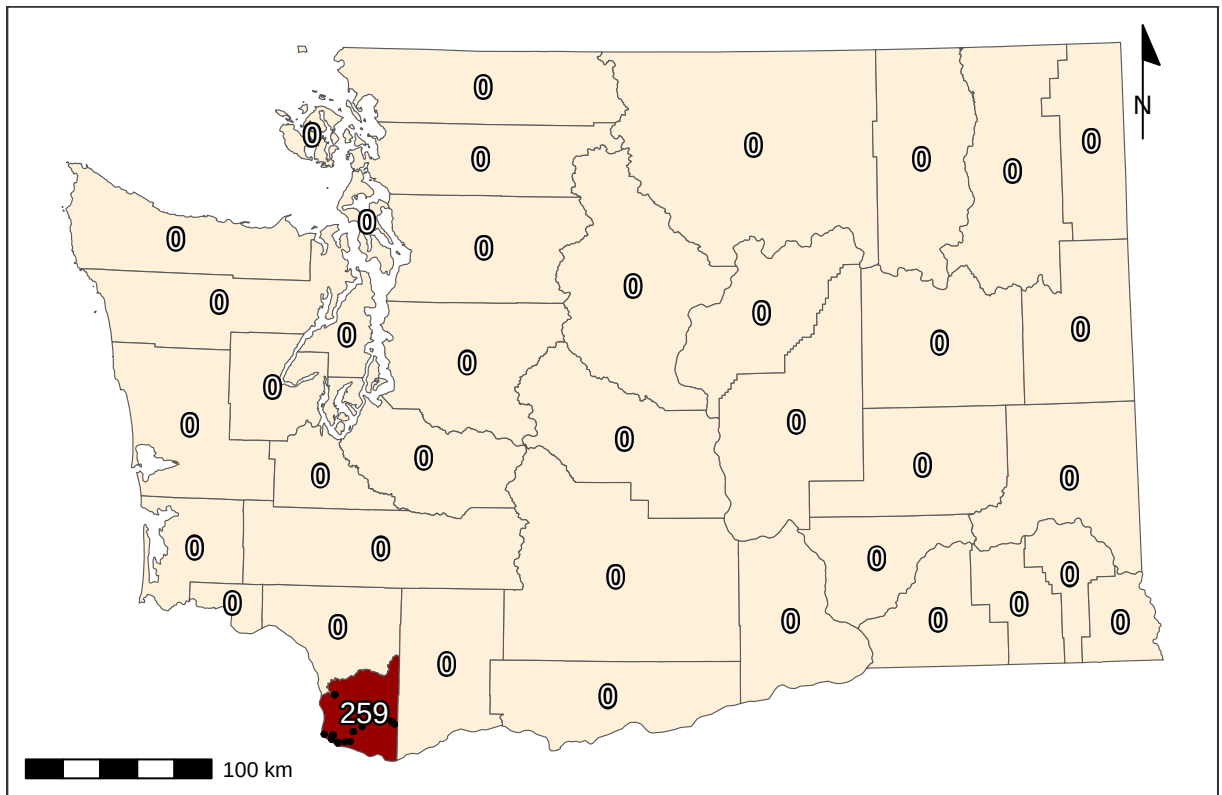


Figure 3: Bee catch locations for Michael O'Loughlin (within Washington), along with total catches per county.

### 3 Total Catches

Volunteers from the Washington Bee Atlas project caught 10006 bees across 35 counties from March 10, 2024 to November 14, 2024, representing 354 unique species and 43 unique genera. The **Nimble Net Kudos** (most specimens collected) goes to Karla Salp, Karen Wright, and Ingrid Carmean, who caught a total of 1689, 1005, and 957 specimens. The *positive* kind of **Darwin Award** (most species collected) goes to Karla Salp, Karen Wright, and Ellery Newcomer, who caught a total of 220, 167, and 153 unique species. Well done!

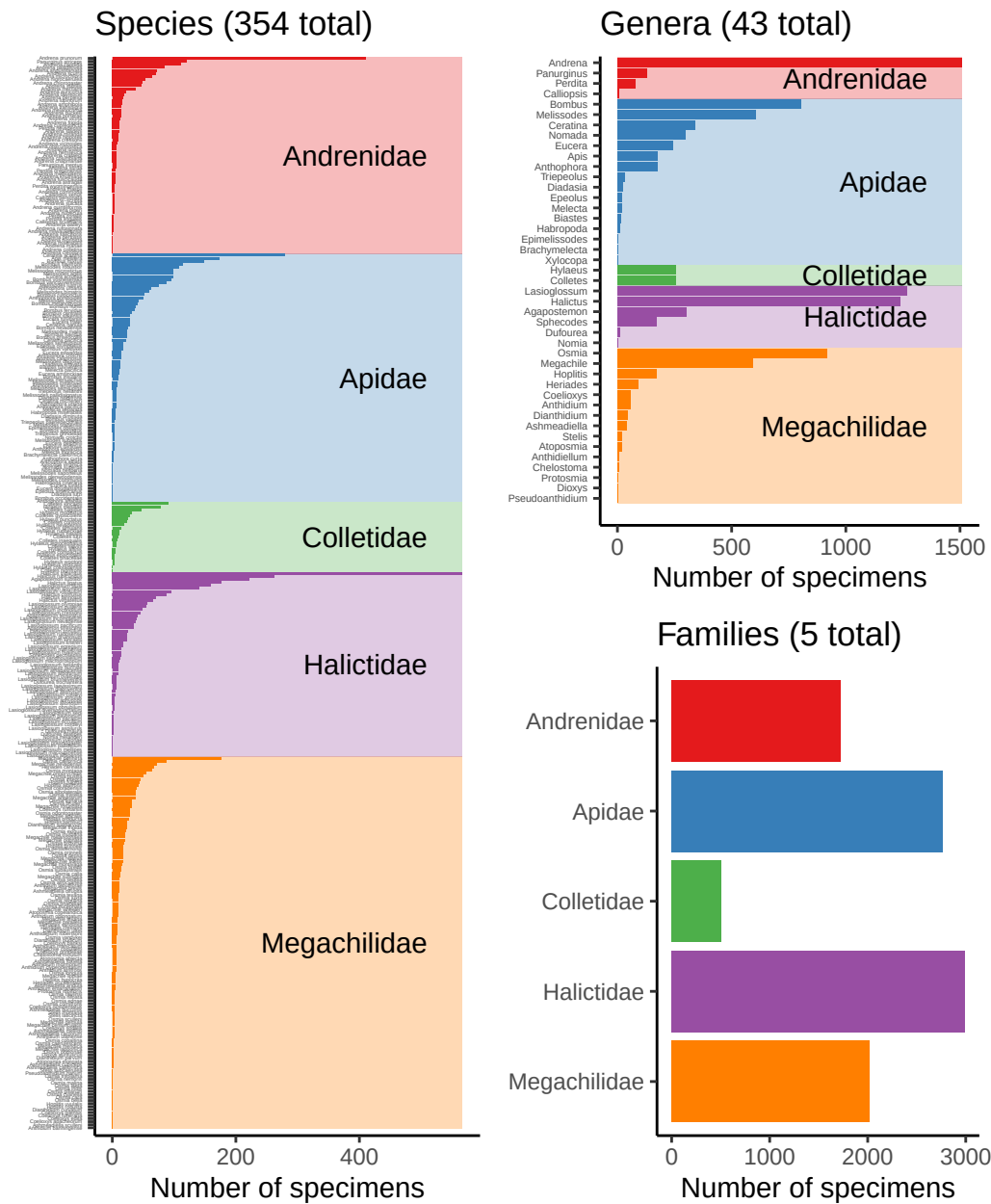


Figure 4: Bees caught by all volunteers, broken down by species, genus, and family.

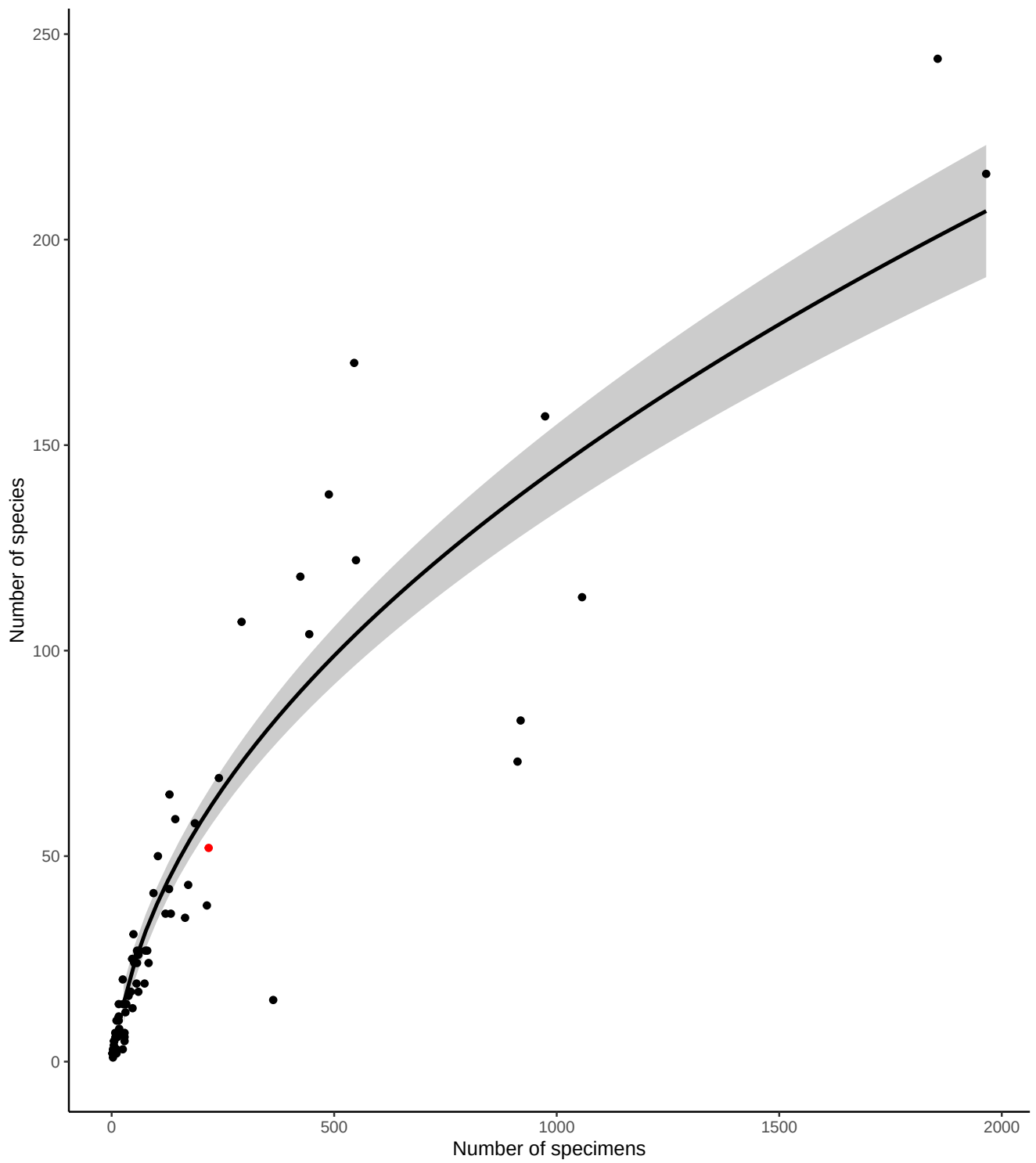


Figure 5: Number of bee specimens and unique bee species caught by all volunteers, with your effort shown in red. This graph should give you an idea of how many specimens you would need to catch to begin seeing rarer bee species.

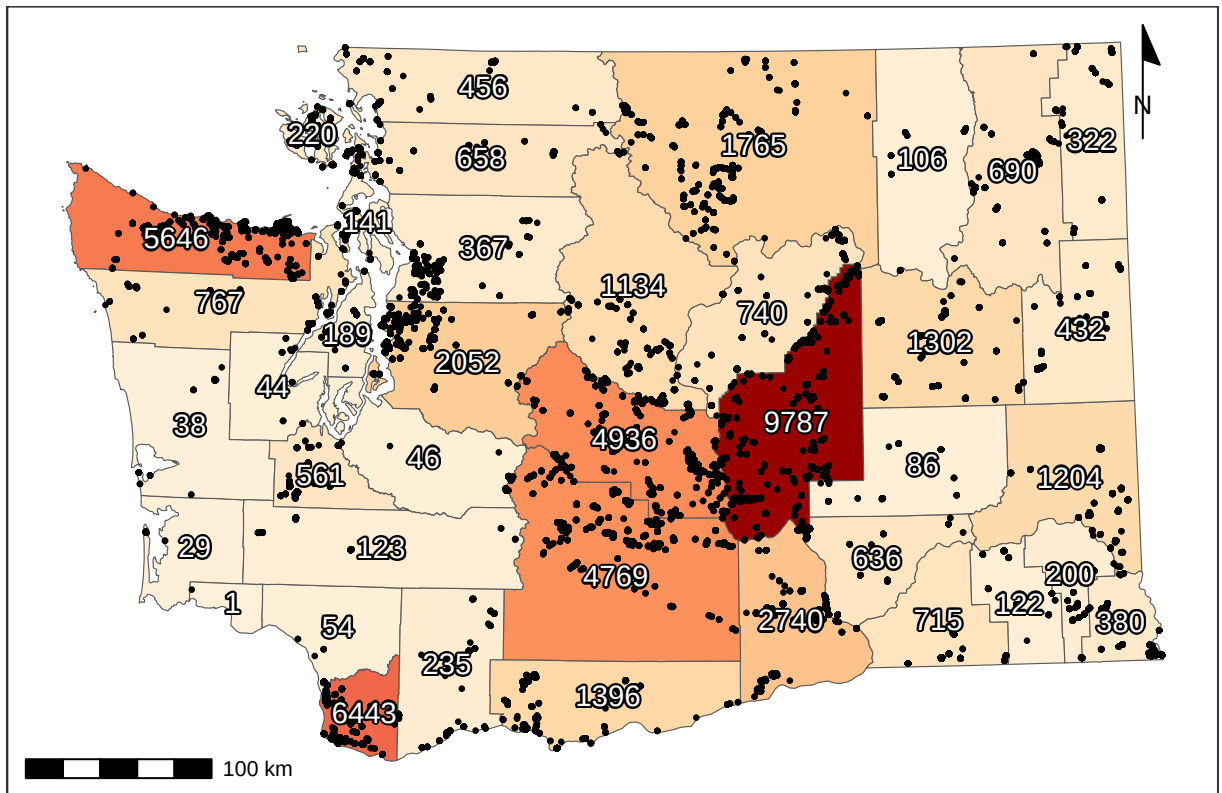


Figure 6: Total specimens caught per county, along with catch location of each specimen (black dots). For genus- and species-specific information for each county, see Tables 3 and 4.

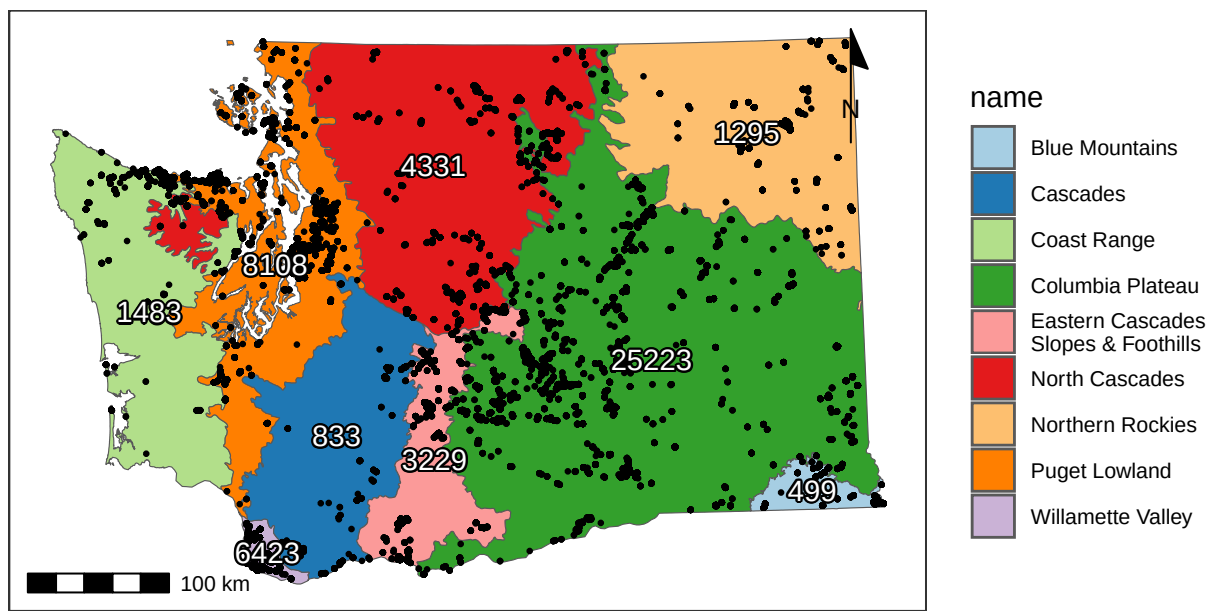


Figure 7: Total catches per ecoregion (Level III), along with catch location of each specimen (black dots).

## 4 Flight Phenology

### 4.1 West of (and including) the Cascade Mountains

Most bees (90%) were caught between April 16 and September 01, but the peak of season (50% of specimens) was from June 11 to July 27.

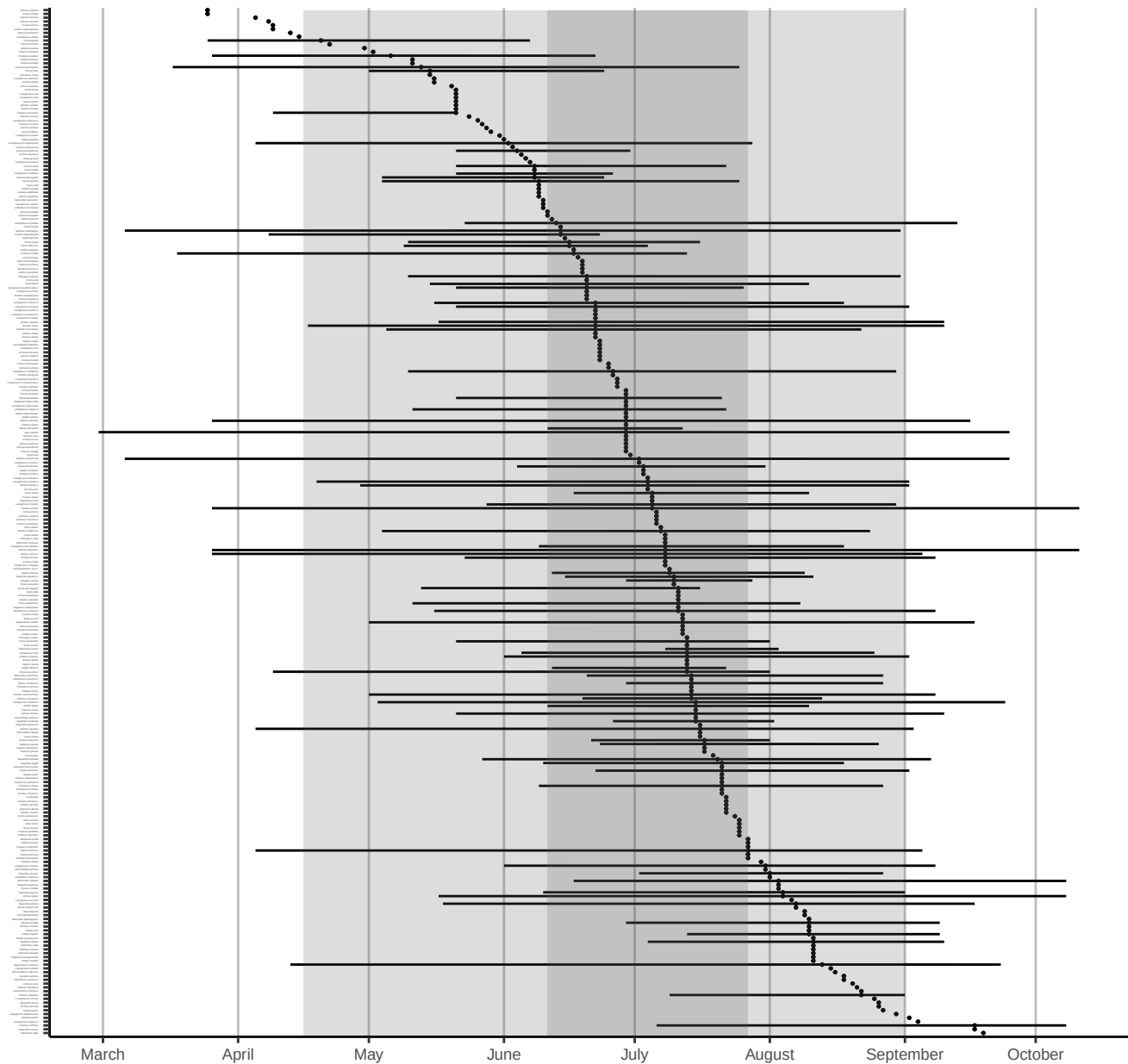


Figure 8: Phenology plot for all bee species caught in or West of the Cascade Mountains, sorted by median abundance times. Percentiles of overall emergence times (50th & 90th) are shown in grey shaded regions. Date ranges for each species (minimum, first, second, third quartiles, and maximum) are shown only for species with >10 specimens.



## 4.2 East of the Cascade Mountains

Most bees (90%) were caught between April 14 and September 18, but the peak of season (50% of specimens) was from May 15 to July 16.

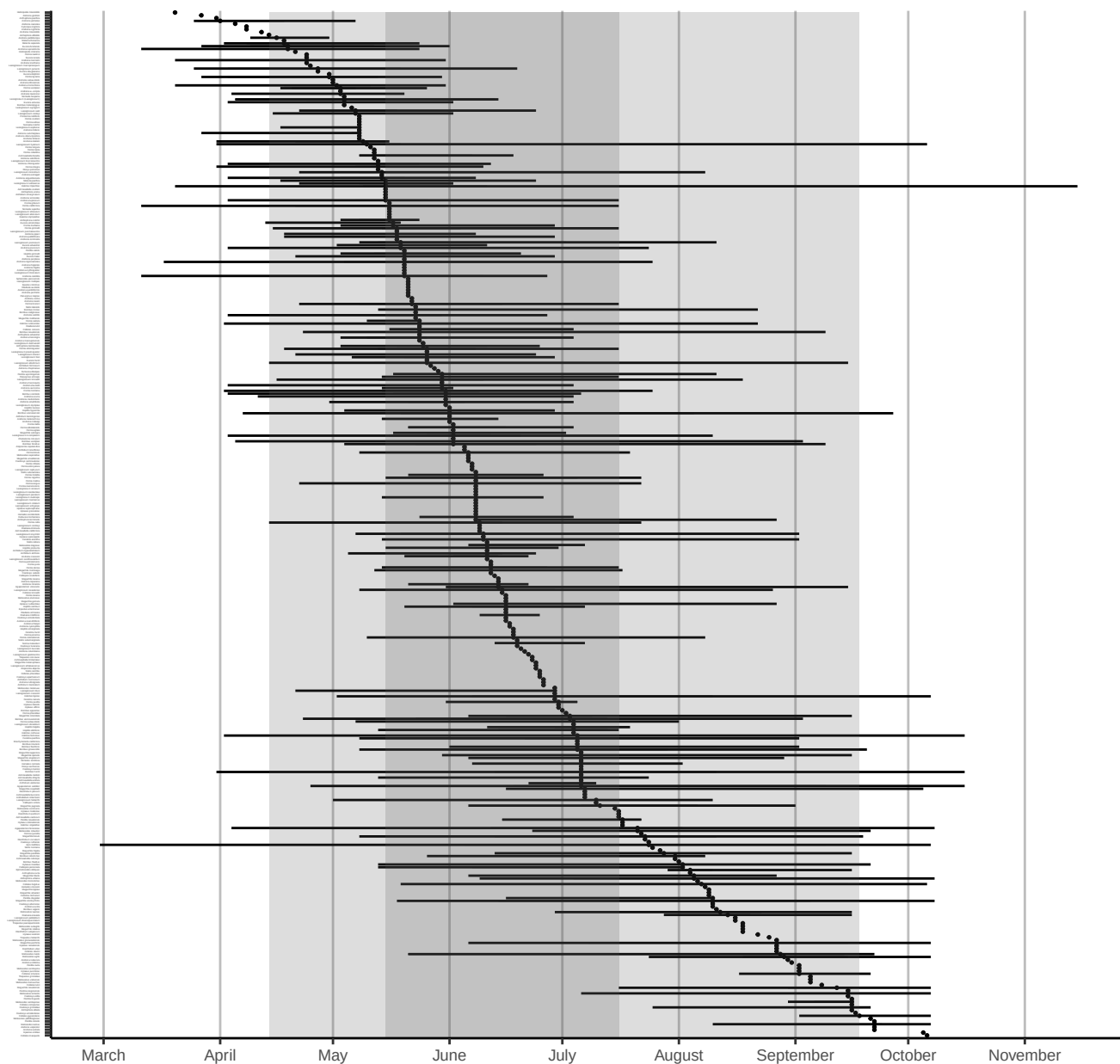


Figure 9: Phenology plot for all bee species caught east of the Cascade Mountains, sorted by median abundance times. Percentiles of overall emergence times (50th & 90th) are shown in grey shaded regions. Date ranges for each species (minimum, first, second, third quartiles, and maximum) are shown only for species with >10 specimens.

## 5 Plant genera

Volunteers collected specimens from a total of 371 unique flower genera, with most volunteers sampling from 12 flower genera (median value). The **Flower Power Kudos** (most sampled flower genera) goes to Karla Salp, Karen Wright, and Nancy Carlson, who collected bees from a total of 170, 155, and 118 genera of flowers. Well done!

The flower genera that had the most specimens caught on them were *Grindelia*, *Ericameria*, and *Lomatium*, which yielded a total of 1095, 731, and 717 specimens. The flower genera that were popular with the most species of bees were *Erigeron*, *Phacelia*, and *Cirsium*, hosting a total of 101, 98, and 91 unique bee species. See Tables 1 and 2 for more details.

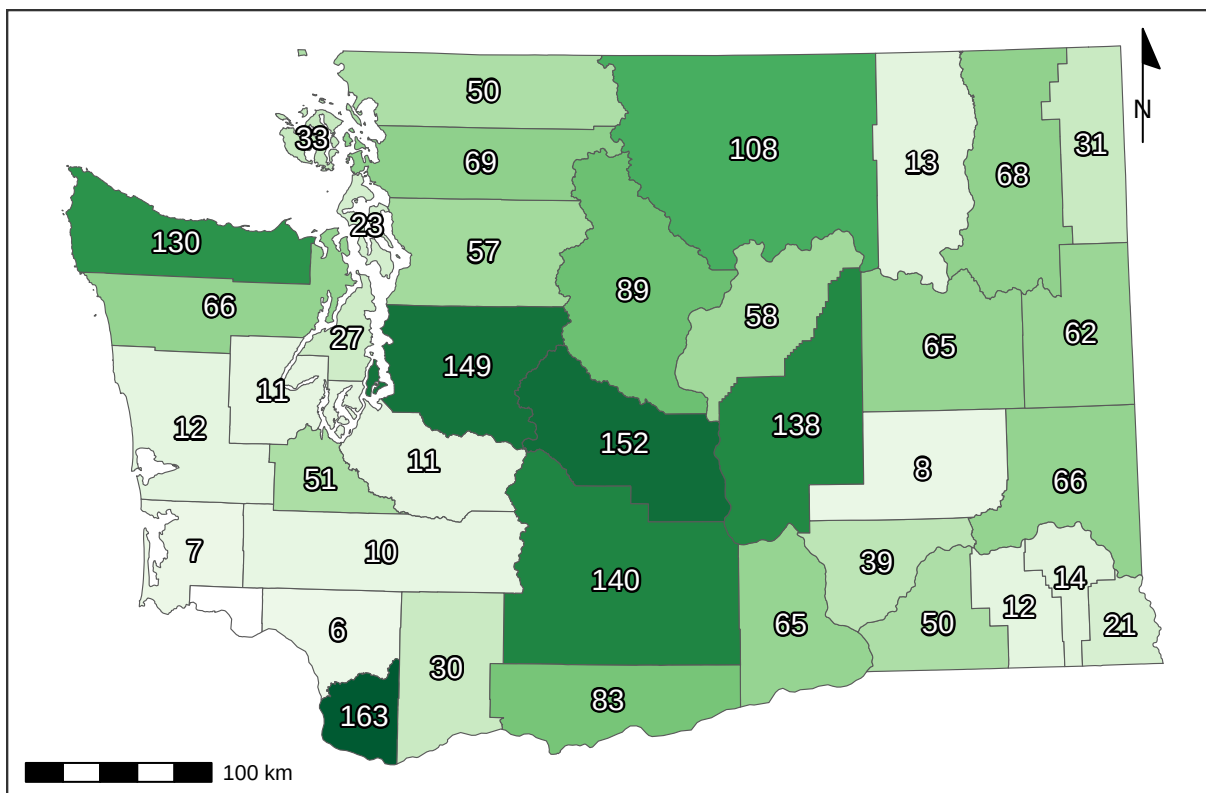


Figure 10: Recorded number of flower genera per county.

Table 1: Number of bee specimens collected from each plant genus. Plants with few records are great targets for future sampling.

| Genus                 | Count | Genus                | Count | Genus                 | Count | Genus                   | Count | Genus                 | Count | Genus                   | Count | Genus                | Count |
|-----------------------|-------|----------------------|-------|-----------------------|-------|-------------------------|-------|-----------------------|-------|-------------------------|-------|----------------------|-------|
| <i>Grindelia</i>      | 1095  | <i>Amsinckia</i>     | 97    | <i>Delphinium</i>     | 35    | <i>Angelica</i>         | 18    | <i>Armeria</i>        | 8     | <i>Anthemis</i>         | 4     | <i>Abelia</i>        | 2     |
| <i>Ericameria</i>     | 731   | <i>Berberis</i>      | 97    | <i>Malva</i>          | 35    | <i>Acmispon</i>         | 17    | <i>Coriandrum</i>     | 8     | <i>Artemisia</i>        | 3     | <i>Androsace</i>     | 2     |
| <i>Lomatium</i>       | 717   | <i>Dieteria</i>      | 94    | <i>Nemophila</i>      | 34    | <i>Euphorbia</i>        | 17    | <i>Cuphea</i>         | 8     | <i>Asparagus</i>        | 3     | <i>Arenaria</i>      | 2     |
| <i>Hypochaeris</i>    | 655   | <i>Eschscholzia</i>  | 94    | <i>Syringa</i>        | 34    | <i>Cacaliopsis</i>      | 16    | <i>Cytisus</i>        | 8     | <i>Barbarea</i>         | 3     | <i>Cardamine</i>     | 2     |
| <i>Phacelia</i>       | 624   | <i>Fragaria</i>      | 93    | <i>Drymocalis</i>     | 33    | <i>Lewisia</i>          | 16    | <i>Dipsacus</i>       | 8     | <i>Dimorphotheca</i>    | 3     | <i>Corydalis</i>     | 2     |
| <i>Helianthus</i>     | 490   | <i>Origanum</i>      | 91    | <i>Descraineria</i>   | 32    | <i>Campanula</i>        | 15    | <i>Downingia</i>      | 8     | <i>Eremogone</i>        | 3     | <i>Cota</i>          | 2     |
| <i>Balsamorhiza</i>   | 471   | <i>Clarkia</i>       | 89    | <i>Malus</i>          | 32    | <i>Draba</i>            | 15    | <i>Hesperochiron</i>  | 8     | <i>Galium</i>           | 3     | <i>Crocasmia</i>     | 2     |
| <i>Citrus</i>         | 450   | <i>Medicago</i>      | 88    | <i>Crataegus</i>      | 31    | <i>Hackelia</i>         | 15    | <i>Aquilegia</i>      | 7     | <i>Gayophytum</i>       | 3     | <i>Crocus</i>        | 2     |
| <i>Erigeron</i>       | 446   | <i>Ceanothus</i>     | 85    | <i>Doellingeria</i>   | 31    | <i>Heterotheca</i>      | 15    | <i>Erica</i>          | 7     | <i>Ceanothus</i>        | 3     | <i>Achlys</i>        | 1     |
| <i>Solidago</i>       | 429   | <i>Daucus</i>        | 83    | <i>Myosotis</i>       | 31    | <i>Lonicera</i>         | 15    | <i>Fritillaria</i>    | 7     | <i>Hibiscus</i>         | 3     | <i>Adelphia</i>      | 1     |
| <i>Penstemon</i>      | 414   | <i>Sphaeralcea</i>   | 82    | <i>Tunacetum</i>      | 31    | <i>Oenothera</i>        | 15    | <i>Viburnum</i>       | 7     | <i>Impatiens</i>        | 3     | <i>Albizia</i>       | 1     |
| <i>Leucanthemum</i>   | 413   | <i>Lathyrus</i>      | 78    | <i>Bellis</i>         | 30    | <i>Perideridia</i>      | 15    | <i>Amelanchier</i>    | 6     | <i>Lapsana</i>          | 3     | <i>Anthozanthum</i>  | 1     |
| <i>Sisymbrium</i>     | 408   | <i>Pediocactus</i>   | 78    | <i>Cleome</i>         | 30    | <i>Ratibida</i>         | 15    | <i>Anchusa</i>        | 6     | <i>Liatris</i>          | 3     | <i>Antirrhinum</i>   | 1     |
| <i>Eriogonum</i>      | 381   | <i>Epilobium</i>     | 77    | <i>Hieracium</i>      | 30    | <i>Antennaria</i>       | 14    | <i>Cynara</i>         | 6     | <i>Lobularia</i>        | 3     | <i>Bellardia</i>     | 1     |
| <i>Lupinus</i>        | 372   | <i>Rudbeckia</i>     | 77    | <i>Rorippa</i>        | 30    | <i>Crocidium</i>        | 14    | <i>Cynoglossum</i>    | 6     | <i>Nothochelone</i>     | 3     | <i>Boechera</i>      | 1     |
| <i>Potentilla</i>     | 320   | <i>Monardella</i>    | 74    | <i>Cerastium</i>      | 29    | <i>Frangula</i>         | 14    | <i>Limnanthes</i>     | 6     | <i>Oenanthe</i>         | 3     | <i>Buglossoides</i>  | 1     |
| <i>Rubus</i>          | 285   | <i>Physocarpus</i>   | 71    | <i>Collinsia</i>      | 29    | <i>Hydrophyllum</i>     | 14    | <i>Luetkea</i>        | 6     | <i>Oreocarya</i>        | 3     | <i>Calystegia</i>    | 1     |
| <i>Rosa</i>           | 269   | <i>Camassia</i>      | 70    | <i>Lactuca</i>        | 29    | <i>Plantago</i>         | 14    | <i>Microseris</i>     | 6     | <i>Physaria</i>         | 3     | <i>Capsella</i>      | 1     |
| <i>Symphoricarpos</i> | 265   | <i>Spiraea</i>       | 70    | <i>Olaynia</i>        | 29    | <i>Borago</i>           | 13    | <i>Nigella</i>        | 6     | <i>Pieris</i>           | 3     | <i>Carduus</i>       | 1     |
| <i>Taraxacum</i>      | 258   | <i>Calochortus</i>   | 69    | <i>Clematis</i>       | 28    | <i>Heleum</i>           | 13    | <i>Onopordum</i>      | 6     | <i>Primula</i>          | 3     | <i>Cercis</i>        | 1     |
| <i>Madia</i>          | 256   | <i>Nepeta</i>        | 68    | <i>Acer</i>           | 27    | <i>Bistorta</i>         | 12    | <i>Phoenixaulis</i>   | 6     | <i>Pseudognaphalium</i> | 3     | <i>Chamaecrista</i>  | 1     |
| <i>Holodiscus</i>     | 247   | <i>Coreopsis</i>     | 67    | <i>Canadanthus</i>    | 27    | <i>Cucurbita</i>        | 12    | <i>Robinia</i>        | 6     | <i>Sagittaria</i>       | 3     | <i>Comandra</i>      | 1     |
| <i>Achillea</i>       | 246   | <i>Agastache</i>     | 66    | <i>Sonchus</i>        | 27    | <i>Heuchera</i>         | 12    | <i>Ageratina</i>      | 5     | <i>Saponaria</i>        | 3     | <i>Cotoneaster</i>   | 1     |
| <i>Asclepias</i>      | 228   | <i>Hypericum</i>     | 60    | <i>Sphaerophysa</i>   | 26    | <i>Persicaria</i>       | 12    | <i>Ajuga</i>          | 5     | <i>Silene</i>           | 3     | <i>Cryptantha</i>    | 1     |
| <i>Lotus</i>          | 224   | <i>Convolvulus</i>   | 59    | <i>Linum</i>          | 25    | <i>Spergularia</i>      | 12    | <i>Ammi</i>           | 5     | <i>Tellima</i>          | 3     | <i>Cucumis</i>       | 1     |
| <i>Trifolium</i>      | 218   | <i>Cichorium</i>     | 55    | <i>Wyethia</i>        | 25    | <i>Vitex</i>            | 12    | <i>Arctostaphylos</i> | 5     | <i>Thalictrum</i>       | 3     | <i>Dicentra</i>      | 1     |
| <i>Crepis</i>         | 217   | <i>Veronica</i>      | 55    | <i>Aruncus</i>        | 24    | <i>Centaurium</i>       | 11    | <i>Bassia</i>         | 5     | <i>Thermopsis</i>       | 3     | <i>Echinops</i>      | 1     |
| <i>Brassica</i>       | 213   | <i>Arnica</i>        | 54    | <i>Chondrilla</i>     | 24    | <i>Chorispura</i>       | 11    | <i>Cymopterus</i>     | 5     | <i>Visnaga</i>          | 3     | <i>Erythronium</i>   | 1     |
| <i>Centauria</i>      | 210   | <i>Digitalis</i>     | 54    | <i>Lithospermum</i>   | 24    | <i>Cosmos</i>           | 11    | <i>Helopsis</i>       | 5     | <i>Dalea</i>            | 2     | <i>Gnaphalium</i>    | 1     |
| <i>Prunus</i>         | 202   | <i>Linaria</i>       | 53    | <i>Anethum</i>        | 23    | <i>Dahlia</i>           | 11    | <i>Hydrangea</i>      | 5     | <i>Eurybia</i>          | 2     | <i>Holcus</i>        | 1     |
| <i>Salix</i>          | 200   | <i>Chaenactis</i>    | 52    | <i>Purshia</i>        | 23    | <i>Micranthes</i>       | 11    | <i>Melissa</i>        | 5     | <i>Fagopyrum</i>        | 2     | <i>Hylotelephium</i> | 1     |
| <i>Ribes</i>          | 188   | <i>Nothocalais</i>   | 51    | <i>Hieracium</i>      | 22    | <i>Pseudotsuga</i>      | 11    | <i>Mycelis</i>        | 5     | <i>Fothergilla</i>      | 2     | <i>Hymenopappus</i>  | 1     |
| <i>Ranunculus</i>     | 184   | <i>Sedum</i>         | 51    | <i>Mentzelia</i>      | 22    | <i>Pyrocampa</i>        | 11    | <i>Narcissus</i>      | 5     | <i>Hesperis</i>         | 2     | <i>Hyssopus</i>      | 1     |
| <i>Anaphalis</i>      | 179   | <i>Euthamia</i>      | 47    | <i>Thelypodium</i>    | 22    | <i>Valerianella</i>     | 11    | <i>Petasites</i>      | 5     | <i>Ionactis</i>         | 2     | <i>Ilex</i>          | 1     |
| <i>Senecio</i>        | 176   | <i>Cornus</i>        | 44    | <i>Triteleia</i>      | 22    | <i>Asperugo</i>         | 10    | <i>Phedimus</i>       | 5     | <i>Kalmia</i>           | 2     | <i>Ipomopsis</i>     | 1     |
| <i>Chrysothamnus</i>  | 175   | <i>Helianthella</i>  | 44    | <i>Papaver</i>        | 21    | <i>Berteroa</i>         | 10    | <i>Sorbus</i>         | 5     | <i>Ladcania</i>         | 2     | <i>Isocoma</i>       | 1     |
| <i>Melilotus</i>      | 167   | <i>Lavandula</i>     | 44    | <i>Polemonium</i>     | 21    | <i>Erodium</i>          | 10    | <i>Argentina</i>      | 4     | <i>Ligusticum</i>       | 2     | <i>Lamprocapnos</i>  | 1     |
| <i>Gaillardia</i>     | 154   | <i>Mentha</i>        | 44    | <i>Erythranthe</i>    | 20    | <i>Lamium</i>           | 10    | <i>Cistus</i>         | 4     | <i>Lycium</i>           | 2     | <i>Lomelosia</i>     | 1     |
| <i>Chamaenerion</i>   | 147   | <i>Plagiobothrys</i> | 44    | <i>Frasera</i>        | 20    | <i>Lygodesmia</i>       | 10    | <i>Dichelostemma</i>  | 4     | <i>Mertensia</i>        | 2     | <i>Lycopus</i>       | 1     |
| <i>Philadelphus</i>   | 146   | <i>Plectritis</i>    | 43    | <i>Iris</i>           | 20    | <i>Marrubium</i>        | 10    | <i>Echinacea</i>      | 4     | <i>Montia</i>           | 2     | <i>Matricaria</i>    | 1     |
| <i>Albium</i>         | 144   | <i>Pastinaca</i>     | 42    | <i>Oemeria</i>        | 20    | <i>Monarda</i>          | 10    | <i>Elaeagnus</i>      | 4     | <i>Nasturtium</i>       | 2     | <i>Muscari</i>       | 1     |
| <i>Lepidium</i>       | 134   | <i>Rhododendron</i>  | 42    | <i>Phlox</i>          | 20    | <i>Solanum</i>          | 10    | <i>Eutrochium</i>     | 4     | <i>Noccea</i>           | 2     | <i>Nicotiana</i>     | 1     |
| <i>Symphoricarpos</i> | 132   | <i>Vaccinium</i>     | 42    | <i>Rhus</i>           | 20    | <i>Tragopogon</i>       | 10    | <i>Glycyrrhiza</i>    | 4     | <i>Osmia</i>            | 2     | <i>Oxalis</i>        | 1     |
| <i>Vicia</i>          | 120   | <i>Lithophragma</i>  | 40    | <i>Toxicoscordion</i> | 20    | <i>Fallopia</i>         | 9     | <i>Iberis</i>         | 4     | <i>Ozomelis</i>         | 2     | <i>Phalaris</i>      | 1     |
| <i>Cilia</i>          | 119   | <i>Erysimum</i>      | 39    | <i>Abronia</i>        | 19    | <i>Pyrrus</i>           | 9     | <i>Ivesia</i>         | 4     | <i>Pedicularis</i>      | 2     | <i>Pisum</i>         | 1     |
| <i>Salvia</i>         | 117   | <i>Nestodus</i>      | 39    | <i>Anthriscus</i>     | 19    | <i>Rhaphiolepis</i>     | 9     | <i>Machaeranthera</i> | 4     | <i>Petroselinum</i>     | 2     | <i>Polygonum</i>     | 1     |
| <i>Jacobaea</i>       | 116   | <i>Calendula</i>     | 38    | <i>Castilleja</i>     | 19    | <i>Tripleurospermum</i> | 9     | <i>Microsteris</i>    | 4     | <i>Pilosella</i>        | 2     | <i>Salsola</i>       | 1     |
| <i>Eriophyllum</i>    | 111   | <i>Dasiphora</i>     | 38    | <i>Gaultheria</i>     | 19    | <i>Leontodon</i>        | 8     | <i>Mutarda</i>        | 4     | <i>Poa</i>              | 2     | <i>Sambucus</i>      | 1     |
| <i>Astragalus</i>     | 104   | <i>Sidalcea</i>      | 38    | <i>Thymus</i>         | 19    | <i>Levistica</i>        | 8     | <i>Opuntia</i>        | 4     | <i>Pyraantha</i>        | 2     | <i>Santolina</i>     | 1     |
| <i>Apocynum</i>       | 102   | <i>Cakile</i>        | 37    | <i>Collomia</i>       | 18    | <i>Paeonia</i>          | 8     | <i>Petrosedum</i>     | 4     | <i>Satureja</i>         | 2     | <i>Sisyrinchium</i>  | 1     |
| <i>Geranium</i>       | 99    | <i>Claytonia</i>     | 37    | <i>Oxytropis</i>      | 18    | <i>Scrophularia</i>     | 8     | <i>Stellaria</i>      | 4     | <i>Symphytum</i>        | 2     | <i>Urtica</i>        | 1     |
| <i>Ilamna</i>         | 99    | <i>Prunella</i>      | 35    | <i>Stephanomeria</i>  | 18    | <i>Stachys</i>          | 8     | <i>Tolmiea</i>        | 4     | <i>Viola</i>            | 2     | <i>Veratrum</i>      | 1     |
| <i>Lythrum</i>        | 97    | <i>Rhaponticum</i>   | 35    | <i>Tagetes</i>        | 18    | <i>Valeriana</i>        | 8     | <i>Verbena</i>        | 4     | <i>Zinnia</i>           | 2     | <i>Xerophyllum</i>   | 1     |

Table 2: Number of bee species collected from each plant genus

| Genus                 | Count | Genus                | Count | Genus                | Count | Genus                   | Count | Genus                   | Count | Genus               | Count | Genus                | Count |
|-----------------------|-------|----------------------|-------|----------------------|-------|-------------------------|-------|-------------------------|-------|---------------------|-------|----------------------|-------|
| <i>Erigeron</i>       | 101   | <i>Coreopsis</i>     | 31    | <i>Bellis</i>        | 15    | <i>Antennaria</i>       | 8     | <i>Ageratina</i>        | 4     | <i>Argentina</i>    | 2     | <i>Abelia</i>        | 1     |
| <i>Phacelia</i>       | 98    | <i>Agastache</i>     | 29    | <i>Crataegus</i>     | 14    | <i>Aruncus</i>          | 8     | <i>Ajuga</i>            | 4     | <i>Artemisia</i>    | 2     | <i>Achlys</i>        | 1     |
| <i>Cirsium</i>        | 91    | <i>Eschscholzia</i>  | 29    | <i>Heracleum</i>     | 14    | <i>Berteroa</i>         | 8     | <i>Ammi</i>             | 4     | <i>Borago</i>       | 2     | <i>Adelphia</i>      | 1     |
| <i>Grindelia</i>      | 91    | <i>Monardella</i>    | 29    | <i>Hieracium</i>     | 14    | <i>Cacaliopsis</i>      | 8     | <i>Cistus</i>           | 4     | <i>Cardamine</i>    | 2     | <i>Albizia</i>       | 1     |
| <i>Eriogonum</i>      | 86    | <i>Berberis</i>      | 28    | <i>Malus</i>         | 14    | <i>Canadanthus</i>      | 8     | <i>Cosmos</i>           | 4     | <i>Cota</i>         | 2     | <i>Androsace</i>     | 1     |
| <i>Penstemon</i>      | 85    | <i>Chrysothamnus</i> | 28    | <i>Malva</i>         | 14    | <i>Cleome</i>           | 8     | <i>Cuphea</i>           | 4     | <i>Crocosmia</i>    | 2     | <i>Anthozanthum</i>  | 1     |
| <i>Solidago</i>       | 82    | <i>Convolvulus</i>   | 28    | <i>Tanacetum</i>     | 14    | <i>Erysimum</i>         | 8     | <i>Cymopterus</i>       | 4     | <i>Dalea</i>        | 2     | <i>Antirrhinum</i>   | 1     |
| <i>Crepis</i>         | 75    | <i>Dieteria</i>      | 28    | <i>Lithospermum</i>  | 13    | <i>Heuchera</i>         | 8     | <i>Dahlia</i>           | 4     | <i>Eurybia</i>      | 2     | <i>Arenaria</i>      | 1     |
| <i>Hypochaeris</i>    | 75    | <i>Fragaria</i>      | 28    | <i>Parashia</i>      | 13    | <i>Hydrophyllum</i>     | 8     | <i>Dichlostemma</i>     | 4     | <i>Galium</i>       | 2     | <i>Bellardia</i>     | 1     |
| <i>Lupinus</i>        | 75    | <i>Calochortus</i>   | 27    | <i>Sonchus</i>       | 13    | <i>Anethum</i>          | 7     | <i>Echinacea</i>        | 4     | <i>Geum</i>         | 2     | <i>Boechera</i>      | 1     |
| <i>Leucanthemum</i>   | 69    | <i>Chaenactis</i>    | 27    | <i>Collomia</i>      | 12    | <i>Anthriscus</i>       | 7     | <i>Eutrochium</i>       | 4     | <i>Glycyrrhiza</i>  | 2     | <i>Buglossoides</i>  | 1     |
| <i>Rosa</i>           | 69    | <i>Lathyrus</i>      | 27    | <i>Doellingeria</i>  | 12    | <i>Campanula</i>        | 7     | <i>Frangula</i>         | 4     | <i>Hesperis</i>     | 2     | <i>Calystegia</i>    | 1     |
| <i>Potentilla</i>     | 67    | <i>Madia</i>         | 26    | <i>Hackelia</i>      | 12    | <i>Centaurium</i>       | 7     | <i>Fritillaria</i>      | 4     | <i>Iberis</i>       | 2     | <i>Capsella</i>      | 1     |
| <i>Sisymbrium</i>     | 64    | <i>Medicago</i>      | 26    | <i>Lactuca</i>       | 12    | <i>Helenium</i>         | 7     | <i>Heliopsis</i>        | 4     | <i>Impatiens</i>    | 2     | <i>Carduus</i>       | 1     |
| <i>Achillea</i>       | 63    | <i>Hypericum</i>     | 25    | <i>Plantago</i>      | 12    | <i>Heterotheca</i>      | 7     | <i>Hydrangea</i>        | 4     | <i>Ionactis</i>     | 2     | <i>Cercis</i>        | 1     |
| <i>Ericameria</i>     | 63    | <i>Rudbeckia</i>     | 25    | <i>Syringa</i>       | 12    | <i>Micranthes</i>       | 7     | <i>Ivesia</i>           | 4     | <i>Kalmia</i>       | 2     | <i>Chamaecrista</i>  | 1     |
| <i>Lomatium</i>       | 63    | <i>Arnica</i>        | 24    | <i>Triteleia</i>     | 12    | <i>Perideridia</i>      | 7     | <i>Machaeranthera</i>   | 4     | <i>Levisticum</i>   | 2     | <i>Comandra</i>      | 1     |
| <i>Balsamorhiza</i>   | 62    | <i>Lythrum</i>       | 24    | <i>Vaccinium</i>     | 12    | <i>Rorippa</i>          | 7     | <i>Microsteris</i>      | 4     | <i>Liatris</i>      | 2     | <i>Corydalis</i>     | 1     |
| <i>Rubus</i>          | 62    | <i>Nepeta</i>        | 24    | <i>Wyethia</i>       | 12    | <i>Stephanomeria</i>    | 7     | <i>Mycelis</i>          | 4     | <i>Ligusticum</i>   | 2     | <i>Cotoneaster</i>   | 1     |
| <i>Centauria</i>      | 60    | <i>Cichorium</i>     | 23    | <i>Collinsia</i>     | 11    | <i>Abronia</i>          | 6     | <i>Narcissus</i>        | 4     | <i>Limnanthes</i>   | 2     | <i>Crocus</i>        | 1     |
| <i>Chamaenerion</i>   | 59    | <i>Clematis</i>      | 23    | <i>Lonicera</i>      | 11    | <i>Amelanchier</i>      | 6     | <i>Oenleria</i>         | 4     | <i>Labularia</i>    | 2     | <i>Cryptantha</i>    | 1     |
| <i>Allium</i>         | 58    | <i>Epilobium</i>     | 23    | <i>Mentzelia</i>     | 11    | <i>Aquilegia</i>        | 6     | <i>Onopordum</i>        | 4     | <i>Lycium</i>       | 2     | <i>Cucumis</i>       | 1     |
| <i>Senecio</i>        | 51    | <i>Claytonia</i>     | 22    | <i>Olsynium</i>      | 11    | <i>Armeria</i>          | 6     | <i>Phedimus</i>         | 4     | <i>Montia</i>       | 2     | <i>Dicentra</i>      | 1     |
| <i>Helianthus</i>     | 50    | <i>Euthamia</i>      | 22    | <i>Sphaerophysa</i>  | 11    | <i>Castilleja</i>       | 6     | <i>Stachys</i>          | 4     | <i>Nasturtium</i>   | 2     | <i>Downingia</i>     | 1     |
| <i>Holodiscus</i>     | 50    | <i>Ilama</i>         | 21    | <i>Thelypodium</i>   | 11    | <i>Crocidium</i>        | 6     | <i>Toxicoscordion</i>   | 4     | <i>Nigella</i>      | 2     | <i>Echinops</i>      | 1     |
| <i>Melilotus</i>      | 50    | <i>Linaria</i>       | 21    | <i>Thymus</i>        | 11    | <i>Cucurbita</i>        | 6     | <i>Anthemis</i>         | 3     | <i>Nothochelone</i> | 2     | <i>Erythronium</i>   | 1     |
| <i>Prunus</i>         | 50    | <i>Nothocalais</i>   | 21    | <i>Acer</i>          | 10    | <i>Erica</i>            | 6     | <i>Arctostaphylos</i>   | 3     | <i>Opuntia</i>      | 2     | <i>Fagopyrum</i>     | 1     |
| <i>Trifolium</i>      | 50    | <i>Sedum</i>         | 21    | <i>Chondrilla</i>    | 10    | <i>Fullopia</i>         | 6     | <i>Asparagus</i>        | 3     | <i>Osmia</i>        | 2     | <i>Fothergilla</i>   | 1     |
| <i>Lotus</i>          | 49    | <i>Philadelphus</i>  | 20    | <i>Descurainia</i>   | 10    | <i>Hesperochiron</i>    | 6     | <i>Barbarea</i>         | 3     | <i>Oxytropis</i>    | 2     | <i>Gnaphalium</i>    | 1     |
| <i>Taraxacum</i>      | 49    | <i>Physocarpus</i>   | 20    | <i>Digitalis</i>     | 10    | <i>Leontodon</i>        | 6     | <i>Bassia</i>           | 3     | <i>Pedicularis</i>  | 2     | <i>Holcus</i>        | 1     |
| <i>Anaphalis</i>      | 45    | <i>Sphaeralcea</i>   | 20    | <i>Euphorbia</i>     | 10    | <i>Monarda</i>          | 6     | <i>Cynara</i>           | 3     | <i>Petroselinum</i> | 2     | <i>Hylotelephium</i> | 1     |
| <i>Gaillardia</i>     | 45    | <i>Camassia</i>      | 19    | <i>Lewisia</i>       | 10    | <i>Persicaria</i>       | 6     | <i>Dimorphotheca</i>    | 3     | <i>Physaria</i>     | 2     | <i>Hymenopappus</i>  | 1     |
| <i>Salvia</i>         | 45    | <i>Cornus</i>        | 19    | <i>Oenothera</i>     | 10    | <i>Rhaphiolepis</i>     | 6     | <i>Elaeagnus</i>        | 3     | <i>Pieris</i>       | 2     | <i>Hyssopus</i>      | 1     |
| <i>Brassica</i>       | 43    | <i>Plectritis</i>    | 19    | <i>Plagiobothrys</i> | 10    | <i>Spergularia</i>      | 6     | <i>Eremogone</i>        | 3     | <i>Pyraecantha</i>  | 2     | <i>Ilex</i>          | 1     |
| <i>Ranunculus</i>     | 43    | <i>Veronica</i>      | 19    | <i>Polemonium</i>    | 10    | <i>Tripleurospermum</i> | 6     | <i>Gayophytum</i>       | 3     | <i>Stellaria</i>    | 2     | <i>Ipomopsis</i>     | 1     |
| <i>Symphotrichum</i>  | 43    | <i>Daucus</i>        | 18    | <i>Pseudotsuga</i>   | 10    | <i>Valerianella</i>     | 6     | <i>Hibiscus</i>         | 3     | <i>Symphytum</i>    | 2     | <i>Isocoma</i>       | 1     |
| <i>Vicia</i>          | 43    | <i>Delphinium</i>    | 18    | <i>Acmispon</i>      | 9     | <i>Anchusa</i>          | 5     | <i>Lamium</i>           | 3     | <i>Thermopsis</i>   | 2     | <i>Ladania</i>       | 1     |
| <i>Lepidium</i>       | 42    | <i>Drymocallis</i>   | 18    | <i>Angelica</i>      | 9     | <i>Asperugo</i>         | 5     | <i>Lapsana</i>          | 3     | <i>Viola</i>        | 2     | <i>Lamprocapnos</i>  | 1     |
| <i>Salix</i>          | 38    | <i>Mentha</i>        | 18    | <i>Bistorta</i>      | 9     | <i>Coriandrum</i>       | 5     | <i>Luetkea</i>          | 3     | <i>Polygonum</i>    | 1     | <i>Lomelosia</i>     | 1     |
| <i>Symphoricarpos</i> | 38    | <i>Nastotus</i>      | 18    | <i>Cerastium</i>     | 9     | <i>Cynoglossum</i>      | 5     | <i>Mutarda</i>          | 3     | <i>Salsola</i>      | 1     | <i>Lycopus</i>       | 1     |
| <i>Asclepias</i>      | 37    | <i>Prunella</i>      | 18    | <i>Chorispora</i>    | 9     | <i>Cytisus</i>          | 5     | <i>Oenanthe</i>         | 3     | <i>Sambucus</i>     | 1     | <i>Matricaria</i>    | 1     |
| <i>Amsinckia</i>      | 35    | <i>Erythranthe</i>   | 17    | <i>Erodium</i>       | 9     | <i>Dipsacus</i>         | 5     | <i>Oreocarya</i>        | 3     | <i>Santolina</i>    | 1     | <i>Melissa</i>       | 1     |
| <i>Apocynum</i>       | 35    | <i>Helianthella</i>  | 17    | <i>Frasera</i>       | 9     | <i>Draba</i>            | 5     | <i>Phoenicautis</i>     | 3     | <i>Satureja</i>     | 1     | <i>Mertensia</i>     | 1     |
| <i>Eriophyllum</i>    | 35    | <i>Linum</i>         | 17    | <i>Iris</i>          | 9     | <i>Gaultheria</i>       | 5     | <i>Primula</i>          | 3     | <i>Sisyrinchium</i> | 1     | <i>Muscari</i>       | 1     |
| <i>Geranium</i>       | 34    | <i>Rhaponticum</i>   | 17    | <i>Nemophila</i>     | 9     | <i>Microseris</i>       | 5     | <i>Pseudognaphalium</i> | 3     | <i>Tellima</i>      | 1     | <i>Nicotiana</i>     | 1     |
| <i>Jacobaea</i>       | 34    | <i>Cakile</i>        | 16    | <i>Pastinaca</i>     | 9     | <i>Paeonia</i>          | 5     | <i>Pyrus</i>            | 3     | <i>Thalictrum</i>   | 1     | <i>Noccea</i>        | 1     |
| <i>Ribes</i>          | 34    | <i>Pedicularis</i>   | 16    | <i>Lithophragma</i>  | 8     | <i>Petasites</i>        | 5     | <i>Robinia</i>          | 3     | <i>Tobimica</i>     | 1     | <i>Oxalis</i>        | 1     |
| <i>Ceanothus</i>      | 33    | <i>Sidalcea</i>      | 16    | <i>Lygodesmia</i>    | 8     | <i>Phlox</i>            | 5     | <i>Sagittaria</i>       | 3     | <i>Urtica</i>       | 1     | <i>Ozomelis</i>      | 1     |
| <i>Clarkia</i>        | 33    | <i>Calendula</i>     | 15    | <i>Marrubium</i>     | 8     | <i>Pyrrocoma</i>        | 5     | <i>Saponaria</i>        | 3     | <i>Veratrum</i>     | 1     | <i>Petrosedum</i>    | 1     |
| <i>Origanum</i>       | 33    | <i>Dasiphora</i>     | 15    | <i>Papaver</i>       | 8     | <i>Solanum</i>          | 5     | <i>Scrophularia</i>     | 3     | <i>Verbena</i>      | 1     | <i>Phalaris</i>      | 1     |
| <i>Spiraea</i>        | 33    | <i>Lavandula</i>     | 15    | <i>Ratibida</i>      | 8     | <i>Tragopogon</i>       | 5     | <i>Silene</i>           | 3     | <i>Visnaga</i>      | 1     | <i>Pilosella</i>     | 1     |
| <i>Astragalus</i>     | 32    | <i>Myosotis</i>      | 15    | <i>Rhus</i>          | 8     | <i>Valeriana</i>        | 4     | <i>Sorbus</i>           | 3     | <i>Xerophyllum</i>  | 1     | <i>Pisum</i>         | 1     |
| <i>Gilia</i>          | 32    | <i>Rhododendron</i>  | 15    | <i>Tagetes</i>       | 8     | <i>Viburnum</i>         | 4     | <i>Vitex</i>            | 3     | <i>Zinnia</i>       | 1     | <i>Poa</i>           | 1     |





|                          | Adams | Asotin | Benton | Chelan | Cllallam | Clark | Columbia | Cowlitz | Douglas | Ferry | Franklin | Garfield | Grant | Grays Harbor | Inland | Jefferson | King | Kitsap | Kittitas | Klickitat | Lewis | Lincoln | Mason | Okanogan | Pacific | Pend Oreille | Pierce | San Juan | Skagit | Skamania | Snohomish | Spokane | Stevens | Thurston | Wahkiakum | Walla Walla | Whitcom | Whitman | Yakima | TOTAL |
|--------------------------|-------|--------|--------|--------|----------|-------|----------|---------|---------|-------|----------|----------|-------|--------------|--------|-----------|------|--------|----------|-----------|-------|---------|-------|----------|---------|--------------|--------|----------|--------|----------|-----------|---------|---------|----------|-----------|-------------|---------|---------|--------|-------|
| Andrena vicinaoides      | 0     | 0      | 0      | 1      | 0        | 0     | 0        | 0       | 0       | 0     | 0        | 0        | 1     | 0            | 0      | 0         | 1    | 0      | 4        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 1        | 0         | 0           | 0       | 4       | 12     |       |
| Andrena vulpcolor        | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 0       | 0     | 0        | 0        | 1     | 0            | 0      | 0         | 0    | 0      | 0        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 1       |        |       |
| Andrena w-scripta        | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 0       | 0     | 0        | 0        | 0     | 0            | 0      | 0         | 0    | 0      | 2        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 1           | 0       | 3       |        |       |
| Andrena walleyi          | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 0       | 0     | 0        | 0        | 0     | 0            | 0      | 0         | 0    | 0      | 2        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 2       |        |       |
| Andrena washingtoni      | 0     | 0      | 0      | 1      | 0        | 0     | 0        | 0       | 0       | 0     | 0        | 0        | 0     | 0            | 0      | 0         | 0    | 0      | 0        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 1       |        |       |
| Anthididulum robertsoni  | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 0       | 2     | 0        | 5        | 0     | 0            | 0      | 0         | 0    | 0      | 2        | 0         | 0     | 0       | 0     | 1        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 1       | 11     |       |
| Anthidium atrifrons      | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 0       | 0     | 2        | 3        | 0     | 0            | 0      | 0         | 0    | 0      | 7        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 7       | 19     |       |
| Anthidium banningsense   | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 0       | 0     | 0        | 4        | 0     | 0            | 0      | 0         | 0    | 0      | 2        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 1       | 7      |       |
| Anthidium clypeodentatum | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 3       | 0     | 0        | 0        | 4     | 0            | 0      | 0         | 0    | 0      | 0        | 0         | 0     | 1       | 0     | 2        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 0       | 10     |       |
| Anthidium emarginatum    | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 0       | 2     | 0        | 0        | 0     | 0            | 0      | 0         | 0    | 0      | 1        | 0         | 0     | 1       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 1       | 5      |       |
| Anthidium formosum       | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 0       | 0     | 0        | 1        | 0     | 0            | 0      | 0         | 0    | 0      | 0        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 1       | 2      |       |
| Anthidium manicatum      | 0     | 0      | 2      | 0      | 4        | 9     | 0        | 0       | 0       | 0     | 1        | 0        | 1     | 1            | 0      | 1         | 4    | 4      | 0        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 1        | 0      | 0        | 1         | 3       | 0       | 1        | 0         | 0           | 0       | 1       | 34     |       |
| Anthidium mormonum       | 0     | 0      | 0      | 0      | 1        | 0     | 0        | 0       | 0       | 0     | 0        | 0        | 1     | 0            | 0      | 0         | 0    | 0      | 1        | 1         | 0     | 0       | 0     | 4        | 0       | 0            | 0      | 0        | 0      | 0        | 0         | 2       | 1       | 0        | 0         | 0           | 0       | 4       | 15     |       |
| Anthidium oblongatum     | 0     | 0      | 0      | 0      | 0        | 3     | 0        | 0       | 0       | 0     | 0        | 0        | 0     | 0            | 0      | 15        | 0    | 0      | 0        | 0         | 0     | 0       | 0     | 0        | 0       | 0            | 0      | 0        | 0      | 0        | 5         | 0       | 0       | 0        | 0         | 0           | 0       | 0       | 23     |       |
| Anthidium tenuiflorae    | 0     | 0      | 0      | 1      | 3        | 0     | 0        | 0       | 0       | 0     | 0        | 0        | 1     | 0            | 0      | 0         | 0    | 0      | 7        | 0         | 0     | 0       | 0     | 1        | 0       | 0            | 1      | 0        | 0      | 0        | 0         | 0       | 0       | 0        | 0         | 0           | 0       | 4       | 18     |       |
| Anthidium utahense       | 0     | 0      | 0      | 0      | 0        | 0     | 0        | 0       | 1       | 1     | 0        | 0        | 5     | 0            |        |           |      |        |          |           |       |         |       |          |         |              |        |          |        |          |           |         |         |          |           |             |         |         |        |       |











Table 5: Your determination accuracy in 2024.

| Taxon                   |
|-------------------------|
| No specimens identified |

## 7 Taxonomic Accuracy, 2024

In 2024, you identified 0 of your 109 specimens to genus level and 0 to species level (see Table 5). In total, volunteers from the Washington Bee Atlas project identified 1.3 % (127) of the 10006 bee specimens to the level of genus, with an average accuracy of 87.4%. Volunteer identifications to species level did not occur or are not included in the data (see Table 6).

Table 6: Determination accuracy for all volunteers in 2024.

| Taxon               | Specimens ID-ed | Correct ID | % Correct |
|---------------------|-----------------|------------|-----------|
| <b>Family</b>       |                 |            |           |
| <i>Andrenidae</i>   | 21              | 16         | 76.2      |
| <i>Apidae</i>       | 52              | 52         | 100.0     |
| <i>Colletidae</i>   | 6               | 3          | 50.0      |
| <i>Halictidae</i>   | 8               | 6          | 75.0      |
| <i>Megachilidae</i> | 43              | 41         | 95.3      |
| <i>TOTAL</i>        | 130             | 118        | 90.8      |
| <b>Genus</b>        |                 |            |           |
| <i>Agapostemon</i>  | 1               | 1          | 100.0     |
| <i>Andrena</i>      | 5               | 0          | 0.0       |
| <i>Anthidium</i>    | 1               | 1          | 100.0     |
| <i>Anthophora</i>   | 6               | 6          | 100.0     |
| <i>Ceratina</i>     | 7               | 7          | 100.0     |
| <i>Coelioxys</i>    | 2               | 2          | 100.0     |
| <i>Colletes</i>     | 4               | 3          | 75.0      |
| <i>Eucera</i>       | 1               | 1          | 100.0     |
| <i>Halictus</i>     | 2               | 0          | 0.0       |
| <i>Hoplitis</i>     | 4               | 4          | 100.0     |
| <i>Hylaeus</i>      | 2               | 0          | 0.0       |
| <i>Lasioglossum</i> | 5               | 4          | 80.0      |
| <i>Megachile</i>    | 18              | 18         | 100.0     |
| <i>Melissodes</i>   | 36              | 33         | 91.7      |
| <i>Osmia</i>        | 17              | 15         | 88.2      |
| <i>Perdita</i>      | 16              | 16         | 100.0     |
| <i>TOTAL</i>        | 127             | 111        | 87.4      |
| <b>Species</b>      |                 |            |           |
| <i>TOTAL</i>        | 0               | 0          | NaN       |

Table 7: Your determination accuracy.

| Taxon                   |
|-------------------------|
| No specimens identified |

## 8 Taxonomic Accuracy, All Years

Over your time in the Atlas you identified 0 of your 259 specimens to genus level and 0 to species level (see Table 7). In total, volunteers from the Washington Bee Atlas project identified 0.6 % (127) of the 22418 bee specimens to the level of genus, with an average accuracy of 87.4%. Volunteer identifications to species level did not occur or are not included in the data (See Table 8).

Table 8: Determination accuracy for all volunteers.

| Taxon               | Specimens ID-ed | Correct ID | % Correct |
|---------------------|-----------------|------------|-----------|
| <b>Family</b>       |                 |            |           |
| <i>Andrenidae</i>   | 21              | 16         | 76.2      |
| <i>Apidae</i>       | 52              | 52         | 100.0     |
| <i>Colletidae</i>   | 6               | 3          | 50.0      |
| <i>Halictidae</i>   | 8               | 6          | 75.0      |
| <i>Megachilidae</i> | 43              | 41         | 95.3      |
| <i>TOTAL</i>        | 130             | 118        | 90.8      |
| <b>Genus</b>        |                 |            |           |
| <i>Agapostemon</i>  | 1               | 1          | 100.0     |
| <i>Andrena</i>      | 5               | 0          | 0.0       |
| <i>Anthidium</i>    | 1               | 1          | 100.0     |
| <i>Anthophora</i>   | 6               | 6          | 100.0     |
| <i>Ceratina</i>     | 7               | 7          | 100.0     |
| <i>Coelioxys</i>    | 2               | 2          | 100.0     |
| <i>Colletes</i>     | 4               | 3          | 75.0      |
| <i>Eucera</i>       | 1               | 1          | 100.0     |
| <i>Halictus</i>     | 2               | 0          | 0.0       |
| <i>Hoplitis</i>     | 4               | 4          | 100.0     |
| <i>Hylaeus</i>      | 2               | 0          | 0.0       |
| <i>Lasioglossum</i> | 5               | 4          | 80.0      |
| <i>Megachile</i>    | 18              | 18         | 100.0     |
| <i>Melissodes</i>   | 36              | 33         | 91.7      |
| <i>Osmia</i>        | 17              | 15         | 88.2      |
| <i>Perdita</i>      | 16              | 16         | 100.0     |
| <i>TOTAL</i>        | 127             | 111        | 87.4      |
| <b>Species</b>      |                 |            |           |
| <i>TOTAL</i>        | 0               | 0          | NaN       |